



Comparative analysis of diffusion mechanisms inside porous media for steam methane reforming over Ni-Al₂O₃ catalyst

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ABSTRACT

The catalyst particles are widely used in steam methane reforming. The main mass transport mechanism inside the catalyst particles is diffusion. Several diffusion mechanisms are used by researchers for the investigations, and no consensus has been reached on which one is more suitable. This study aims to find the appropriate diffusion mechanism for the steam methane reforming. The effect of various diffusion mechanisms on the predictive performances of the steam methane reforming process was investigated for Ni-Al₂O₃ catalyst particles. Four diffusion mechanisms were considered: constant diffusion coefficient, Lewis number correlation, kinetic theory, and Bosanquet approximation. The CH₄ conversion, species and temperature distributions were compared for various operating temperatures, residence time, particle diameters, pressure and H₂O/CH₄ ratio (β). For the similar operating and design parameters, CH₄ conversion obtained for Bosanquet approximation and Lewis number correlation showed close to similar results with average deviation of 2%. Other diffusion models have failed to correctly predict the CH₄ conversion. It was shown that the choice of an appropriate diffusion mechanism is important for the numerical prediction of the steam methane reforming performances.

1. Introduction

Steam methane reforming (SMR) is an essential chemical engineering process used to produce hydrogen and synthesis gas [1,2]. Currently, approximately 95% of hydrogen is generated via the steam reforming of natural gas, which primarily consists of methane [3,4]. Despite SMR being a well-known process for over 100 years, the number of publications on this topic continues to increase [5,6]. Scientists investigate design parameters of the reactor [7], new catalysts [8], new applications [9] and influence of the operating conditions on reaction efficiency [10]. The steam methane reforming process is very complex. It includes endothermic chemical reactions, heat and mass transfer inside the reactor, and complex fluid flow [11]. Steam methane reforming is a catalytic process, and in industrial applications, the porous catalysts are often placed inside the reactor as a packed bed. The temperature and species distributions inside the catalyst particle significantly influence the efficiency of SMR reactions. Therefore, the heat and mass transfer inside porous catalysts is important in overall reformer performance [12].

Publications on steam methane reforming can be broadly categorized into the following subgroups: catalysis, industrial applications,

laboratory experiments, and numerical simulations. The catalyst physical and chemical parameters such as particle size, porosity, and good activity, influence the SMR efficiency significantly [13]. Some authors also investigate the properties of catalyst material including Pt, Yt, Rh, Ru, Pd, Pt, Ni and others [14,15]. Steam methane reforming studies in industrial applications usually aim at hydrogen or syngas production [8,11]. However, there are other applications, for example, waste heat recuperation using endothermic steam methane reforming reactions [16–18]. Industrial processes are hard to study on-site. Therefore, researchers use various models or laboratory-scale experiments. Laboratory experiments are often used to understand fundamental characteristics like reaction kinetics [19]. Various models are used to evaluate the application of new techniques such as electrically-driven heating [20], reactors with a membrane [21], energy storage [22,23] and solar heating [24].

In recent years, numerical simulation of the steam methane reforming process has garnered significant interest among engineers and scientists worldwide [25,26]. Several reasons for this interest are the continual increase in computational power of personal computers and

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the development and enhancement of modern tools for numerical simulations. These advancements enable high-accuracy numerical results that closely align with experimental data [27].

Numerical simulation of steam methane reforming encompasses various processes, including flow dynamics, chemical reactions, heat and mass transfer, as well as diffusion inside and outside of porous catalytic particles [28,29]. Additionally, numerical simulation of steam methane reforming can be classified into two groups: particle-resolved method (PRM) [30–32] and pseudo-homogeneous model (PHM) [33–35]. In PRM, each catalyst particle is represented as a part of the computational domain, with all processes of steam methane reforming considered for each particle and extra-particle space [31]. In PHM, the reaction space of the reformer is presented as a continuum without presence of particles with source terms accounting for the momentum loss [36]. The processes mentioned above are considered for that medium [37,38].

Ullah et al. [37] used PHM of steam methane reformer with membrane system to compare palladium-based membrane with pressure swing absorption technology. Ben-Mansour et al. [21] investigated various heat supply methods in membrane-integrated steam methane reformer using the PHM. Tacchino et al. [39] used the PHM method to study the coupling between flame and steam methane reformer. The authors note that significant parameter variability demonstrates the importance of particle-resolved simulation. The PHM method treats a packed bed as a medium with evenly spread catalysts and calculates average flow velocity and heat transfer. As a result, PHM does not account for local heat and mass transfer phenomena. Therefore, it could not be used to study various particle shapes, intraparticle mass transfer and other local phenomena.

The intra-particle reaction, heat and mass transfer should be accounted for as they significantly influence the macroscopic gradients inside the packed bed. Numerical modeling of SMR reactions using PRM is well-suited for this purpose. Dixon [31] numerically investigated the distributions of temperature, species and reaction rates in industrial-scale packed bed using the PRM approach. The catalyst particles were treated as solid regions and user-defined scalars were used to model intra-particle heat and mass transfer. The strong local gradients of temperature and species mass fractions were observed between neighboring particles and even on the same particle surface. In another study by Dixon et al. [40], the local variations in flow characteristics, heat and mass transfer were compared for particles with various shapes. It was established that the reaction occurs mainly in the thin regions near the catalyst surface. The variations in local flow patterns significantly influenced the local distribution of temperature, species and reaction rate. In these studies, the authors implemented Langmuir–Hinshelwood–Hougen–Watson (LHHW) chemical kinetics for simulations.

Many papers on the numerical simulation of steam methane reforming implement LHHW chemical kinetics described by Xu and Froment [41,42], and Hou and Hughes [43,44]. In their papers [45,46], the authors derived the chemical kinetics model for intrinsic kinetics when the catalyst particles are very small (with an average diameter of about 0.1–0.15 mm). For such diameters, no intra-particle diffusion limitations occur in the steam methane reforming process. However, it is well-known that an increase in catalyst particle diameter leads to increased intra-particle diffusion limitations [47]. Therefore, diffusion is an important process to be modeled for steam methane reforming, especially for industrial-scale applications.

Various diffusion models were applied for the numerical simulation of steam methane reforming. Cherif et al. [48] numerically investigated the influence of packed bed patterns on hydrogen yield. He implemented a constant diffusion coefficient for all species. The packed bed considered in this investigation had a high density (2400 kg/m³), which indicates the presence of very small catalyst particles. In such applications, it is determined that chemical kinetics is intrinsic, and catalyst particles have no diffusion limitations.

In industry, a catalyst particle is usually larger than 0.15 mm and has diffusion limitations. Lu and Nikrityuk [26] considered a 3D numerical model of a steam methane reformer filled with spherical catalysts. The catalyst particle diameter was 25.4 mm, and the authors considered the heat and mass transfer process inside the particle. They have chosen the unity Lewis number assumption to account for the diffusion of the gas species outside the catalyst particle (bulk diffusion). Inside the catalyst, the authors considered both bulk and Knudsen diffusion.

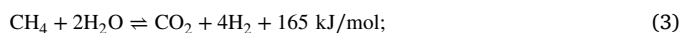
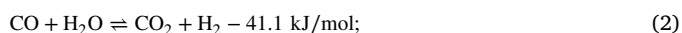
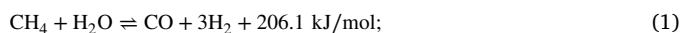
Xiu et al. investigated the diffusion limitations of the catalyst for steam methane reforming reactions in absorption-enhanced reformer [49]. The authors used the Fickian diffusion model to estimate the effectiveness factor — the rate of reaction rate decrease due to the catalyst diffusion limitations. For intra-particle diffusion, they used a binary diffusivity described by Fuller et al. [50]. In that model, the Knudsen diffusivity was not taken into account.

The comparison of the existing diffusion models allows to choose the one with minor deviations and sensible complexity to reduce the simulation time. Solsvik and Jakobsen [51] compared the performance of four diffusion models: Maxwell–Stefan, dusty gas, kinetic theory and Bosanquet approximation for steam methane reforming and methanol synthesis. The authors established that kinetic theory deviates from the Maxwell–Stefan model by several percent. They also remark that these deviations could be caused by numerical error due to high temperature and species gradients. The authors investigated the phenomena for a low temperature of about 800 K, which is less than the typical temperature used for steam methane reforming.

Researchers use various diffusion models for numerical modeling of steam methane reforming, and no consensus has been reached on which diffusion model is more suitable. In general, four diffusion mechanisms applied for numerical simulation of steam methane reforming can be highlighted: constant diffusion coefficient, Lewis number, kinetic theory, and Bosanquet approximation. This article aims to close the knowledge gap on the use of an appropriate diffusion model for steam methane reforming. In this paper, a comparative analysis of diffusion mechanisms inside porous media for steam methane reforming over Ni–Al₂O₃ catalyst was performed. The four aforementioned diffusion mechanisms were tested to understand their effect on calculated performances of steam methane reforming, particularly methane conversion, for various catalyst particle sizes. Moreover, each diffusion model was estimated for a wide range of operating conditions: operating temperature, pressure, steam-to-methane ratio and mixture residence time.

2. Methodology

The steam methane reforming process includes various chemical reactions that convert methane and steam into hydrogen, carbon monoxide, carbon dioxide and carbon formation reactions. Among the different chemical reactions in steam methane reforming, three reactions play a significant role in heat and material balance [52]:



In this paper, the reaction species are represented by five gaseous chemical elements (CH₄, CO, CO₂, H₂O, H₂). The steam methane reforming process is accompanied by other side reactions, e.g., carbon formation and consumption. Carbon formation is essential for steam methane reforming, particularly at elevated temperatures [53]. Existing literature suggests that carbon formation can be managed by enhancing the coking resistance of catalysts and adjusting operational parameters, such as steam-to-methane ratio [54]. In this investigation, the steam-to-methane ratio in all varied feed gas compositions equals or exceeds unity, indicating ample steam for oxidation, effectively preventing solid carbon formation. Consequently, carbon formation was excluded from equilibrium expressions.

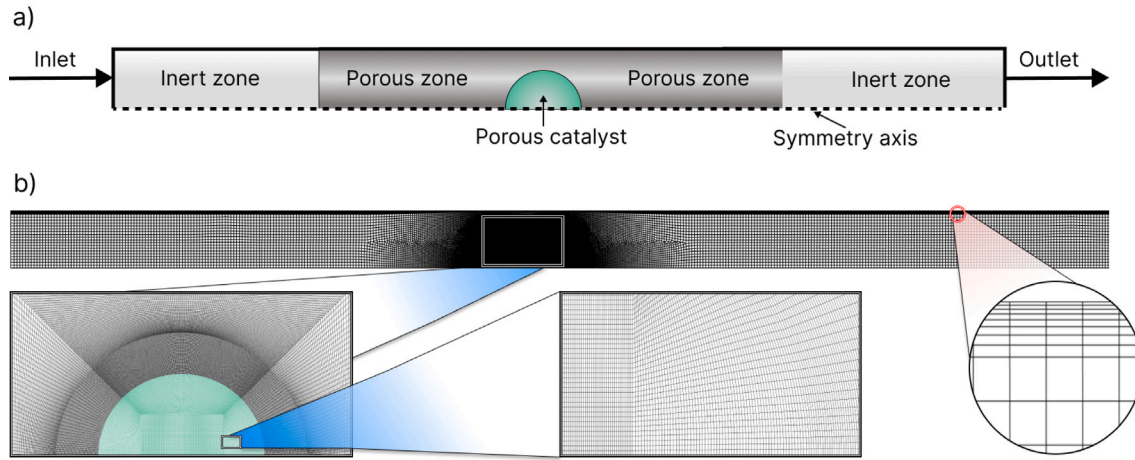


Fig. 1. The computational domain (a) and the mesh structure (b).

Table 1
Influence of mesh elements number on CH₄ conversion and average outlet temperature.

Number of elements	CH ₄ conversion (%)	Temperature (K)
1.21 · 10 ⁴	13.546	998.54
2.34 · 10 ⁴	13.812	998.16
4.16 · 10 ⁴	14.854	997.81
6.41 · 10 ⁴	15.042	997.51
8.25 · 10 ⁴	15.049	997.46
1.37 · 10 ⁵	15.049	997.45

2.1. Geometry and mesh

The numerical simulation of the steam methane reforming process was conducted for a conventional tubular reformer filled with Ni/αAl₂O₃ catalysts. The reformer consists of four zones: the inlet zone, inert inlet and outlet zones are empty tube sections for flow development and vortex dissipation, respectively. The heating zone is filled with inert catalyst particles. These inert catalyst particles provide conditions inside the reaction zone close to isothermal. A porous catalyst particle is placed at the center of the heating zone, as shown in Fig. 1. The catalyst contains 15% NiO on αAl₂O₃ support. This catalyst type is widely adopted in industry, and its properties are thoroughly described by Hou and Hughes [46]. The reformer is a cylindrical tube with a diameter of D=20 mm, the length of each inert zone is 50 mm, and the length of the reaction zone is 200 mm. The wall temperature remains constant along the length of the reaction zone. The computational domain exhibits radial symmetry; thus, a 2D axisymmetric simplification was applied.

The mesh is depicted in Fig. 1b. Quadrilateral elements were selected as the meshing elements. The inflation was created near the reformer wall to capture the laminar sublayer. Special attention was paid to the mesh structure of catalyst particles. The structured mesh with reduced element size was implemented around every catalyst. A mesh independence study was conducted to define the optimal number of elements and is presented below (see Table 1).

2.2. Governing equations

The CFD model equations are associated with particle-resolved computational fluid modeling. The governing equations are based on mass, momentum and energy conservation equations. Inside the catalyst particles the chemical kinetics model equations were solved.

The continuity equation for steady state flow:

$$\nabla \cdot (\rho \vec{v}) = 0 \quad (4)$$

The momentum equation in the general form:

$$\nabla \cdot (\rho \vec{v} \vec{v}) = -\nabla p + \nabla \cdot (\vec{\tau}) + \rho \vec{g} + S_i \quad (5)$$

In these equations, ρ is mixture density; $\vec{\tau}$ is a stress tensor; p is the static pressure; $\rho \vec{g}$ is gravitational body force; S_i is external body force that contains porous media resistance.

The conservation equation of energy is written as:

$$\nabla \cdot \vec{v}(p + \rho \cdot h_f) = \nabla \cdot [k_{eff} \cdot \nabla T - (\sum h_j \cdot \vec{J}_j) + \vec{\tau} \cdot \vec{v}] + S_f^h \quad (6)$$

where h_f is mixture enthalpy; h_j is enthalpy of individual component; k_{eff} is effective thermal conductivity; $\vec{J}_j$ is the diffusion flux of species j ; S_f^h is a source term that includes heat of chemical reactions.

Effective thermal conductivity (k_{eff}) of porous media is derived from:

$$k_{eff} = \epsilon k_f + (1 - \epsilon) k_s \quad (7)$$

where ϵ is porous media porosity; k_f is fluid phase thermal conductivity and k_s is the thermal conductivity of solid material.

The equation for component transportation can be defined as follows:

$$\nabla \cdot (\rho \cdot \vec{v} \cdot Y_i) = -\nabla \cdot \vec{J}_i + R_i \quad (8)$$

where R_i is reaction rate of formation or consumption of i th element; Y_i is the mass fraction of i th element in the gas mixture; $\vec{J}_i$ is diffusion flux of i th element.

The diffusion of i th component in the surrounding flow can be expressed as:

$$\vec{J}_i = - \left(\rho D_{i,m} + \frac{\mu_t}{Sc_t} \right) \nabla Y_i - D_{T,i} \frac{\nabla T}{T} \quad (9)$$

where Sc_t is the turbulent Schmidt number ($\frac{\mu_t}{\rho D_t}$), where μ_t is the turbulent viscosity and D_t is the turbulent diffusivity.

A momentum source term (S_i) was added to momentum conservation equation (Eq. (5)) to account for porous media resistance. The source term includes two components: the inertial loss term and the viscous loss term. The Ergun equation is used to calculate these terms. This semi-empirical relation applies to a wide range of Reynolds numbers and various packed bed types [55,56]. The source term (S_i) caused by inertial resistance and viscous resistance is defined by $1/\alpha$ and C_2 , respectively:

$$S_i = -\frac{\mu}{\alpha} v_i + C_2 \rho |v_i| v_i \quad (10)$$

$$\alpha = \frac{D_p^2}{150} \frac{\epsilon^3}{(1 - \epsilon)^2}; \quad C_2 = \frac{3.5}{D_p} \frac{(1 - \epsilon)}{\epsilon^3} \quad (11)$$

where μ is kinematic viscosity; D_p is the mean particle diameter; ϵ is the packed bed porosity.

2.3. Diffusion mechanisms

The species mass transfer inside the catalyst is strongly limited by diffusion limitations [57]. In turn, species diffusion significantly influences the enthalpy field. It is of great interest to compare the performance of various diffusion models. The investigated diffusion models are presented below:

1. Constant diffusion coefficient
2. Lewis number
3. Kinetic theory
4. Bosanquet approximation

The packed bed is a porous media with various particle sizes. The variety of particle diameters creates long paths of fluid inside the catalyst. The tortuosity factor accounts for this flow obstruction. Therefore, it is included in the diffusion estimation as follows:

$$D_{eff,j} = \frac{\epsilon}{\tau} \overline{D_j} \quad (12)$$

where τ is the tortuosity factor calculated by $1/\epsilon$ [58]; ϵ is porosity of catalyst particle; D_j is species effective diffusion.

2.3.1. Constant diffusion coefficient

The constant diffusion model is a constant value of the effective diffusion. This constant does not account for change of any variables. Therefore, it could be used only in cases with simplified effective diffusion mechanism. The default value proposed by Fluent is as follows [59]:

$$D_{eff,j} = 2.88 \cdot 10^{-5} \text{ m}^2/\text{s} \quad (13)$$

2.3.2. Lewis number correlation

The mass diffusion coefficient for the reaction mixture could be calculated using the Lewis number close to unity. This approach assumes an equal influence of thermal and mass diffusion on species transport. It is quite widespread for numerical simulations [26,48]. The mass diffusion coefficient ($D_{j,m}$) for Lewis number model is written as:

$$Le_j = \frac{k}{\rho c_p D_{j,m}} = 1 \quad (14)$$

$$D_{j,m} = \frac{k_g}{c_p \cdot \rho} \quad (15)$$

where k is thermal conductivity; ρ is density; c_p is isobaric heat capacity; $D_{j,m}$ is mass diffusion coefficient.

2.3.3. Kinetic theory

The kinetic theory model calculates multicomponent diffusion using the species binary diffusion. The diffusion coefficient $D_{i,m}$ of every species i in the mixture is determined as:

$$D_{i,m} = \frac{1 - X_i}{\sum_{j,j \neq i} (X_j/D_{ij})} \quad (16)$$

The binary diffusion $D_{i,j}$ between every pair of species i and j is determined using Chapman-Enskog formula [60]:

$$D_{ij} = 0.00188 \cdot \frac{\left[T^3 \left(\frac{1}{M_{\omega,i}} + \frac{1}{M_{\omega,j}} \right) \right]^{\frac{1}{2}}}{p_{abs} \sigma_{ij}^2 \Omega_D} \quad (17)$$

where p_{abs} is the absolute pressure; $M_{\omega,i}$ and $M_{\omega,j}$ are the molecular mass of component i and j ; σ_{ij} is the arithmetic average of Lennard-Jones collision diameter for species i and j ; Ω_D is the diffusion collision integral, which measures the molecule's interaction in the system. The Ω_D is a function of the quantity T_D^* , where:

$$T_D^* = \frac{T}{(\epsilon/k_B)_{ij}} \quad (18)$$

k_B is the Boltzmann constant, which is defined as the gas constant, R , divided by Avogadro's number. $(\epsilon/k_B)_{ij}$ for the mixture is the geometric average:

$$(\epsilon/k_B)_{ij} = \sqrt{(\epsilon/k_B)_i (\epsilon/k_B)_j} \quad (19)$$

where $(\epsilon/k_B)_i$ is Lennard-Jones potential well depth for species i .

For a binary mixture, the arithmetic average of Lennard-Jones collision diameter for species i and j is calculated as the arithmetic average of the individual σ_s :

$$\sigma_{ij} = \frac{1}{2} (\sigma_i + \sigma_j) \quad (20)$$

The Lennard-Jones potential well depth values $(\epsilon/k_B)_{ij}$ and the Lennard-Jones collision diameter σ_i were taken from the Chemkin database [61] for every species.

2.3.4. Bosanquet approximation

The Bosanquet approximation calculates the mass diffusion coefficient based on species binary diffusion and Knudsen diffusion as shown in Fig. 2.

Knudsen diffusion occurs in porous catalyst particles with relatively small pore diameters. The presence of Knudsen diffusion is determined by the Knudsen number (Kn):

$$Kn = \frac{\lambda}{L} \quad (21)$$

where λ is mean-free-path of a gas molecule and L is a physical length, which in this case is pore diameter d_p .

Mean-free-path of gas molecule λ is written as:

$$\lambda = \frac{\mu}{\rho} \sqrt{\frac{\pi}{2RT}} \quad (22)$$

The Bosanquet approximation is applicable as it falls in the range of Knudsen number $0.01 < Kn < 10$.

The effective binary gas diffusivity coefficient $D_{i,eff}$ is computed as:

$$D_{i,eff} = \frac{\epsilon}{\tau} \frac{1}{1/D_{m,i} + 1/D_{k,i}} \quad (23)$$

where τ is tortuosity and ϵ is pore porosity.

The species diffusion coefficient $D_{m,i}$ is written as:

$$D_{i,m} = \frac{1 - X_i}{\sum_{j,j \neq i} (X_j/D_{ij})} \quad (24)$$

where X_i and X_j are molar fractions of species i and j , respectively.

The binary diffusivity D_{ij} was estimated using the dependence described by Fuller et al. [50]:

$$D_{ij} = \frac{1 \cdot 10^{-3} \cdot T^{1.75} (1/M_i + 1/M_j)^{1/2}}{p \cdot (V_i^{1/3} + V_j^{1/3})^2} \quad (25)$$

where M_i is the species molar weight; and V_i is diffusion volume of species i .

The multi-component Knudsen diffusion coefficient $D_{k,i}$ is written as [62]:

$$D_{k,i} = \frac{d_p}{3} \sqrt{\frac{8RT}{\pi \cdot M_i/2}} \quad (26)$$

where d_p is an average pore diameter.

2.4. Reaction kinetics

One of the most utilized reaction kinetics developed by Hou and Hughes [46] was used for the present investigation. The reaction rates of the chemical reactions Eqs. (1)–(3) are determined based on this kinetic model and calculated as follows:

$$r_1 = \frac{k_1 (P_{CH_4} P_{H_2O}^{0.5} / P_{H_2}^{1.25}) (1 - (P_{CO} P_{H_2}^3 / K_{p1} P_{CH_4} P_{H_2O}))}{(DEN)^2} \quad (27)$$

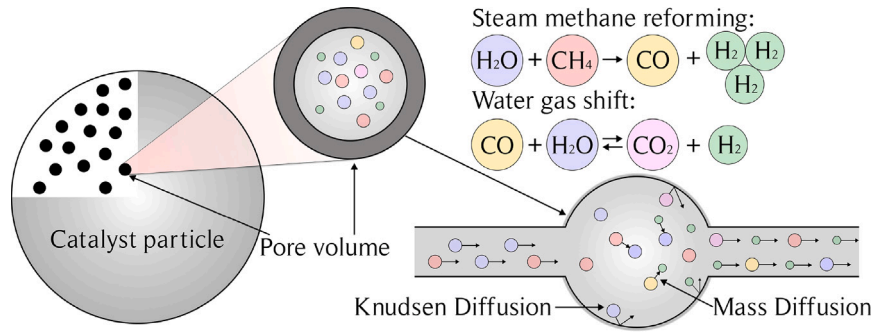


Fig. 2. Diffusion mechanism inside the pore of catalyst particle for Bosanquet approximation.

$$r_2 = \frac{k_2(P_{\text{CO}}P_{\text{H}_2\text{O}}^{0.5}/P_{\text{H}_2}^{0.5})(1 - (P_{\text{CO}_2}P_{\text{H}_2}/K_{p_2}P_{\text{CO}}P_{\text{H}_2\text{O}}))}{(\text{DEN})^2} \quad (28)$$

$$r_3 = \frac{k_3(P_{\text{CH}_4}P_{\text{H}_2\text{O}}/P_{\text{H}_2}^{1.75})(1 - (P_{\text{CO}_2}P_{\text{H}_2}^4/K_{p_3}P_{\text{CH}_4}P_{\text{H}_2\text{O}}^2))}{(\text{DEN})^2} \quad (29)$$

where $k_{1,2,3}$ are reaction rate constants of reaction Eqs. (1)–(3); P_i - partial pressure of species i ($i = \text{CH}_4, \text{CO}, \text{CO}_2, \text{H}_2, \text{H}_2\text{O}$); $K_{p1,2,3}$ are equilibrium constants of reaction Eqs. (1)–(3).

$$\text{DEN} = 1 + K_{\text{CO}}P_{\text{CO}} + K_{\text{H}_2}P_{\text{H}_2}^{0.5} + K_{\text{H}_2\text{O}}(P_{\text{H}_2\text{O}}/P_{\text{H}_2}) \quad (30)$$

where K_i are adsorption constants of species i .

The reaction rate constants of reaction Eqs. (1)–(3) are written as:

$$k_1 = 5.922 \cdot 10^8 \exp\left(\frac{-214200}{8.314T}\right); \quad k_2 = 6.028 \cdot 10^{-4} \exp\left(\frac{19000}{8.314T}\right) \quad (31)$$

$$k_3 = 1.093 \cdot 10^3 \exp\left(\frac{-111800}{8.314T}\right) \quad (32)$$

The adsorption constants for species i are taken as:

$$K_{\text{H}_2\text{O}} = 9.251 \exp\left(\frac{-15900}{8.314T}\right); \quad K_{\text{CO}} = 5.127 \cdot 10^{-13} \exp\left(\frac{140000}{8.314T}\right) \quad (33)$$

$$K_{\text{H}_2} = 5.68 \cdot 10^{-10} \exp\left(\frac{93400}{8.314T}\right) \quad (34)$$

The equilibrium constants for the reaction Eqs. (1)–(3) are:

$$K_1 = 1.198 \cdot 10^{17} \exp\left(\frac{-26830}{8.314T}\right); \quad K_2 = 1.767 \cdot 10^{-2} \exp\left(\frac{4400}{8.314T}\right) \quad (35)$$

$$K_3 = 2.117 \cdot 10^{15} \exp\left(\frac{-22430}{8.314T}\right) \quad (36)$$

Coke formation is an important factor in steam methane reforming. However, the coking could be mitigated using techniques such as improving catalyst coking resistance [63] or increasing the steam-to-methane ratio. Increased steam-to-methane ratio promotes reactions that convert carbon and carbon-forming intermediates back into gaseous products. In the current investigation, the steam-to-methane ratio was always higher than unity, which indicates that the amount of steam was always sufficient for carbon formation prevention. Thus, carbon deposition reactions were not considered in the current investigations.

2.5. Boundary conditions

Numerical calculations were carried out for various operating parameters such as temperature, steam-to-methane ratio (β), operating pressure, and residence time. The residence times were varied from 1500 to 12000 kg s/kmol_{CH₄}. In this paper, the mixture residence time was varied by reaction mixture velocity at the inlet. All the simulations were performed for each investigated diffusion model. The total mass of the catalyst was constant for all cases and equaled 3.16 g. The catalyst particle diameter varied in the range of 3.15–10 mm; e.g., the number

Table 2
Operating and boundary conditions.

Operating parameter	Value	Units
Inlet temperature T_{in}	700–1300	K
Reformer wall temperature $T_{wall} = T_{in}$	700–1300	K
Operating pressure P_{op}	120–1000	kPa
Steam-to-methane ratio β	1.5–5	(–)
Residence time	1500–12000	$\frac{\text{kg s}}{\text{kmol}_{\text{CH}_4}}$
Particle diameter D_p	3.15–10	mm

of particles is 32 for a particle diameter of 3.15 mm and only one particle for a diameter of 10 mm.

Steam methane reforming is an endothermic process with temperature gradients in the reformer. High reaction rates are observed under higher reaction temperatures. The heat was supplied through the wall to bring the reactor conditions closer to isothermal. Moreover, the initial mixture temperature was also equal to the wall temperature. The chemical kinetics developed by Hou and Hughes [46] was based on experiments with additional H_2 supply to the mixture inlet. This was done to suppress the catalyst reoxidation by steam. A constant H_2 mole fraction of 1/7 was supplied on the inlet for all simulations to bring numerical simulations closer to experimental observations.

A pressure-based solver was chosen for the present numerical model. The convergence criteria were set at 10^{-5} for momentum and continuity, 10^{-6} for energy and species. Standard initialization was used for solution initialization. The Second-order Upwind approximation method was chosen for pressure–velocity coupling.

The boundary conditions for the computational domain are listed in Table 2 and expressed as the following:

- Reformer inlet: $T_{in} = T_{in}^{gas}$; $C_i = C_i^{in}$;
 - Reformer outlet: $\frac{\partial C_i}{\partial x} = 0$; $\frac{\partial T^{gas}}{\partial x} = 0$;
- where C_i is the mole concentration of i th element.

3. Results

3.1. Model validation

The numerical model was validated against experimental data obtained by Hou and Hughes [46]. The computational domain for validation purposes was based on geometry described by Hou and Hughes [46]: reformer diameter – 10 mm, and reformer length – 205 mm. The reaction space is filled with $\text{Ni-Al}_2\text{O}_3$ catalysts and inert particles. The catalyst particle diameter was 0.15 mm with a total mass of 0.3 g. The catalyst particle diameter is small. Therefore, reaction kinetics is deemed as intrinsic without the intra-particle diffusion limitation. The pseudo-homogeneous porous media model allows for convenient simulation of the flow through such a packed bed and was implemented in this numerical model. The heat for endothermic reactions was supplied through the wall. The wall temperature was kept constant to sustain the near-isothermal reformer operation. The inlet temperature was the

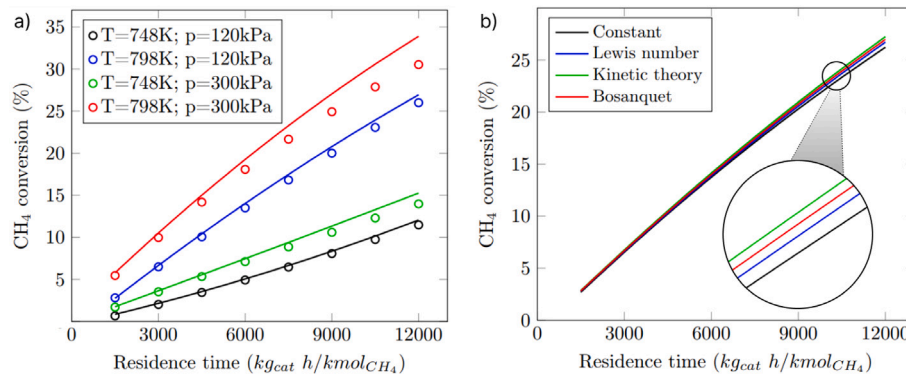


Fig. 3. Validation of numerical simulation (lines) with experimental results (hollow points) from Hou and Hughes [46] (a) and comparison of diffusion models (b).

same as the wall temperature in two cases, 748 K and 798 K. The operating pressure was set at 120 kPa and 300 kPa. The residence time varied from 1500 to 12000 kg s/kmol_{CH₄} due to the various inlet velocity magnitude. The inlet mixture composition ratio was H₂O:CH₄:H₂ = 5:1:1 by the mole fractions.

The validation results are shown in Fig. 3a. The methane conversion from simulations and experiments were compared for various operating conditions. The methane conversion (X_{CH_4}) is calculated as:

$$X_{CH_4} = \frac{X_{in} - X_{out}}{X_{in}}; \quad (37)$$

where X_{in} and X_{out} are the average CH₄ mole fraction on inlet and outlet, respectively.

The numerical simulation results are very close to experimental measurements. The average error is 3.44% for operating pressure of 120 kPa and 5.77% for 300 kPa. The overall average error is below 5.2%, with a maximum deviation of 10.8%. This clearly shows the reliability of the proposed numerical model. Various diffusion mechanisms were also tested for the case with an operating pressure of 120 kPa and temperature of 798 K, presented in Fig. 3b. The discrepancies in CH₄ between mechanisms are pretty low due to the small catalyst particle diameter. The small particle diameter is known to have practically no diffusion limitations. Therefore, the diffusion mechanisms should be investigated on catalyst particles with higher diameters.

3.2. Diffusion model comparison

The contours of CH₄ mole fraction distribution and temperature are shown in Fig. 4a and Fig. 4b. The variables are compared for four diffusion mechanisms under the following operating conditions: particle diameter $D_p = 10$ mm; inlet and wall temperature $T_{in} = 948$ K; residence time 9000 kg s/kmol_{CH₄} and operating pressure $P_{op} = 120$ kPa. The scale values were equal for all cases to better compare the results.

The methane mole fraction decreases after the flow reaches the catalyst particle. The lowest mole fraction is observed inside the catalyst volume for all cases. A sharp mole fraction gradient is also observed near the catalyst surface. The low CH₄ mole fraction inside the porous catalyst could be attributed to considerable local flow resistance. The flow resistance slows the mass transfer significantly. As a result, the mixture residence time inside the catalyst becomes significant, and nearly all available CH₄ is converted. The quantitative comparison of CH₄ conversion between four diffusion mechanisms will be discussed further in this section. The temperature contours (Fig. 4b) show that the kinetic theory diffusion model has the lowest temperature that shows more active heat consumption via endothermic reactions. Lewis number and Bosanquet approximation have similar temperature distributions, and the constant diffusion model has the highest average temperature. It was also observed that a lower CH₄ mole fraction inside the volume of the catalyst corresponds to a higher average temperature.

The effective diffusion coefficients of CH₄ and H₂ for four diffusion mechanisms are shown in Fig. 5. The operating conditions are the same as for Fig. 4. It should be noted that effective diffusion coefficients were calculated for all species: CH₄, H₂, CO, CO₂ and H₂O. However, only CH₄ and H₂ species are shown in Fig. 5 as they are the main reactant and reaction product, respectively. The constant diffusion model generally has the lowest value that stays constant throughout the computational domain. The effective diffusion calculated with the Lewis number is lower outside the catalyst and higher inside it for all species. This behavior is attributed to lower mixture density inside the catalyst as it contains a high H₂ mole fraction. The Kinetic theory predicts diffusion differently for CH₄ and H₂. The effective diffusion coefficient increases inside the catalyst for CH₄ and decreases for H₂, but values vary insignificantly. Bosanquet approximation shows that the diffusion coefficient inside the catalyst particle is much lower than outside. This effect arises due to the implementation of Knudsen diffusion inside the Bosanquet approximation model.

Methane conversion was compared for four diffusion mechanisms under various residence times and particle diameters. The CH₄ conversion for a single particle with particle diameter $D_p = 10$ mm is shown in Fig. 6a and for four particles with diameter $D_p = 6.3$ mm in Fig. 6b. The total mass of particles in case Fig. 6a and Fig. 6b is the same. The wall and mixture inlet temperature is 948K; H₂O:CH₄ = 2:1; operating pressure is 120 kPa.

Methane conversion increases for higher residence time for all cases. This is characteristic behavior as higher residence time provides longer contact of the reaction mixture with the catalyst. Longer contact time allows the reaction to convert more reactants. Diffusion is the main mass transfer mechanism inside a porous catalyst, as its resistance to the flow is relatively high. Therefore, diffusion mechanisms greatly influence the flow residence time.

Bosanquet approximation and unity Lewis number correlation are very close to each other. This indicates that unity Lewis number correlation predicts mass transfer inside catalyst particles similarly to Bosanquet approximation. Constant mass diffusion coefficient significantly diminishes mass transfer, resulting in lower reactant flow inside the catalyst and consequently leading to lower CH₄ conversion. Kinetic theory diffusion mechanism allows the mixture to have practically the same intra-particle mass transfer due to the diffusion as in the flow outside the catalyst particle. This results in higher mass transfer due to the diffusion. Higher mass transfer inside the catalyst decreases diffusion limitation, which elevates CH₄ conversion.

The CH₄ conversion has a strong dependence on particle diameter. Smaller catalyst particles have higher CH₄ conversion for all diffusion models as depicted in Fig. 6b. However, the dependence of CH₄ conversion from residence time has the same behavior as for particle with larger diameter.

As is evident from Fig. 6, smaller particle diameter increases CH₄ conversion but does not change the dependence character of CH₄ conversion from residence time. Therefore, the effect of particle diameter

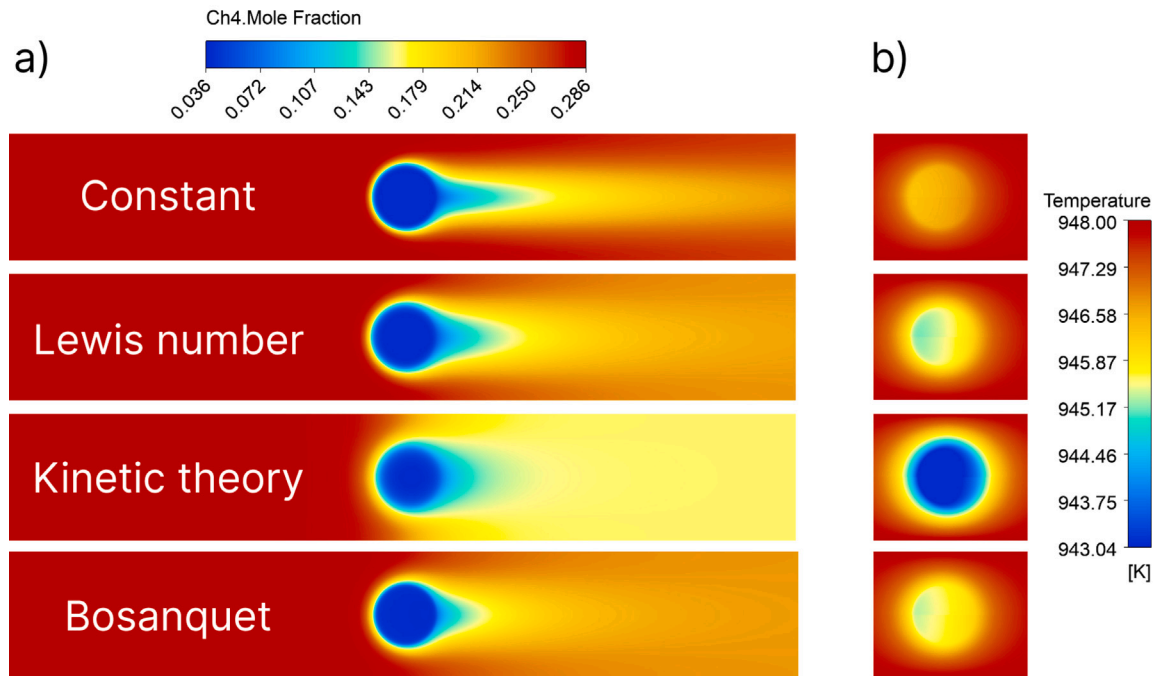


Fig. 4. Distribution of (a) CH₄ molar fraction and (b) temperature inside catalyst particle for various diffusion mechanisms for $D_p = 10$ mm; $T_{in} = 948$ K; residence time 9000 kg s/kmol_{CH₄}.

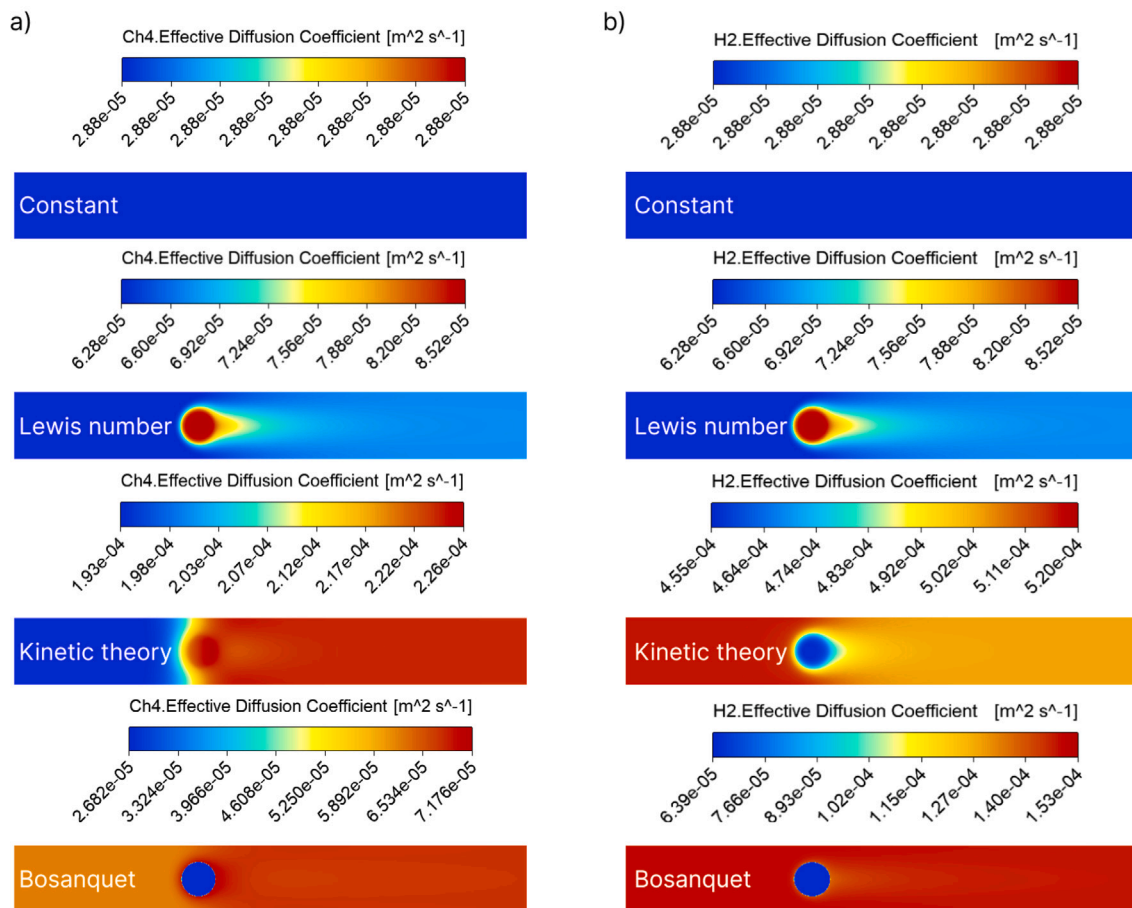


Fig. 5. Effective diffusion coefficient for (a) CH₄ species and (b) H₂ species for $D_p = 10$ mm; $T_{in} = 948$ K; residence time 9000 kg s/kmol_{CH₄}.

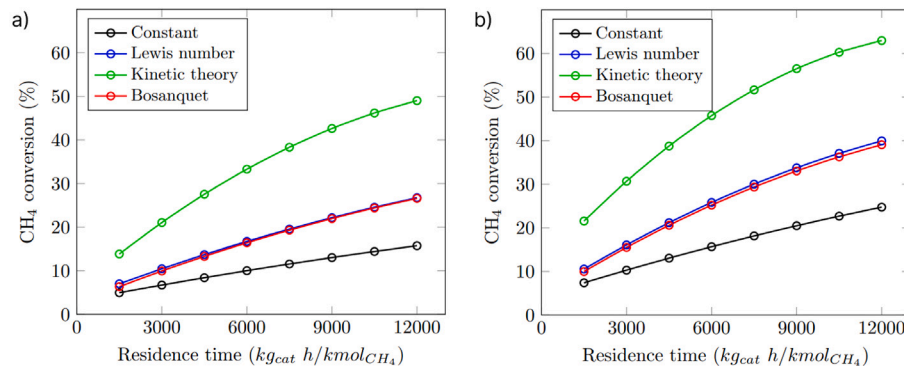


Fig. 6. Methane conversion from residence time for (a) one particle with diameter 10 mm; (b) four particles with diameter 6.3 mm each.

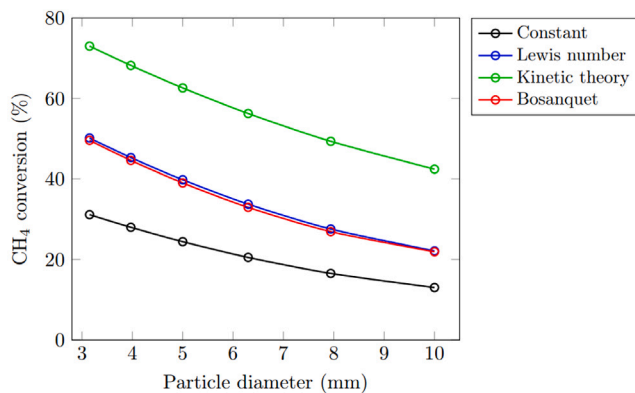


Fig. 7. Methane conversion from particle diameter for $T_{wall} = 948\text{K}$; $\text{H}_2\text{O}:\text{CH}_4 = 2:1$; operating pressure of 120 kPa and residence time of 9000 kg s/kmol_{CH₄}.

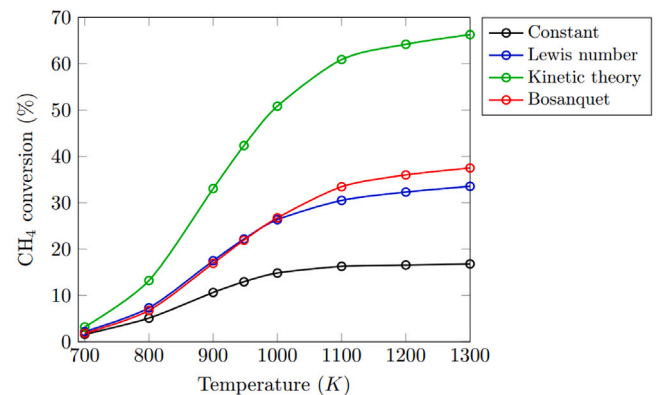


Fig. 8. Methane conversion from reformer temperature for $\text{H}_2\text{O}:\text{CH}_4 = 2:1$; operating pressure of 120 kPa, residence time of 9000 kg s/kmol_{CH₄} and particle diameter 10 mm.

was considered separately for the same set of operating and boundary conditions. The methane conversion with respect to particle diameter is shown in Fig. 7 for inlet and wall temperature of 948K; $\text{H}_2\text{O}:\text{CH}_4 = 2:1$; operating pressure of 120 kPa and residence time of 9000 kg s/kmol_{CH₄}. The case was considered for the following number of particles: 1, 2, 4, 8, 16, and 32. The number of particles varied due to the change in particle diameter. The same total catalyst mass was kept for all cases.

From Fig. 7, it is observed that CH₄ conversion declines with increasing particle diameter. Larger particle diameter has higher diffusion limitations, which hinders the CH₄ conversion. The dependence has the same tendency for all diffusion models. Practically no discrepancies are observed between the unity Lewis number and Bosanquet approximation models. The constant diffusion coefficient underestimates the mass transfer due to the diffusion, and the kinetic theory model significantly overestimates it.

All the considered diffusion models except constant diffusion coefficient calculate effective diffusion with respect to temperature. The CH₄ conversion with respect to inlet and wall temperature is shown in Fig. 8. The residence time of 9000 kg s/kmol_{CH₄}; ratio of $\text{H}_2\text{O}:\text{CH}_4 = 2:1$; operating pressure of 120 kPa and particle diameter of 10 mm were kept the same. The characteristic dependence of CH₄ conversion from temperature is observed for all diffusion models. The CH₄ conversion is very low for 700 K and rapidly increases with higher temperature. After 1100 K the reaction rate approaches its equilibrium and increases insignificantly with temperature. The observed behavior corresponds well to the existing literature [64]. The difference between the diffusion models is similar to the cases presented in Figs. 6 and 7. However, unity Lewis number and Bosanquet approximation diffusion models show various CH₄ conversion for temperatures above 1000 K. It is clearly seen that for temperatures below 1000 K discrepancies between the models are very low. Unity Lewis number model assumes equal

thermal and mass diffusion and this assumption may work improperly for elevated temperatures. The difference between these two models does not exceed 10% for all reformer temperatures.

The operating pressure in the reformer is usually higher than atmospheric. The comparison of CH₄ conversion for various diffusion models under an operating pressure range of 250–1000 kPa is shown in Fig. 9. The reformer temperature is 948 K; residence time is 9000 kg s/kmol_{CH₄}; the ratio of $\text{H}_2\text{O}:\text{CH}_4 = 2:1$, and the particle diameter is 10 mm. Lower methane conversion is observed for higher operating pressure for all diffusion models. The main reason is the suppression of forward reaction, according to Le Chatelier's principle. The kinetic theory diffusion model shows the highest CH₄ conversion and the constant diffusion model shows the lowest. Bosanquet approximation and Lewis number diffusion models have slightly deviating results, and the difference is more pronounced on the lower pressures. Both of these models calculate the effective diffusion coefficient inversely proportional to operating pressure. However, the Bosanquet approximation includes Knudsen diffusion, and that term lowers the influence of binary diffusion, which includes the pressure term. The highest difference between these two models is below 7%.

Steam-to-methane ratio (β) is usually higher than one in industrial applications to prevent carbon deposition on the catalyst surface. In Fig. 10, the CH₄ conversion for various diffusion models is shown with respect to β . The methane conversion decreases with an increase in β for all diffusion models. This agrees with Hou and Hughes [46] as they reported a slower methane disappearance rate with increasing β . The dependencies have a similar structure for all models, with the kinetic theory model predicting the highest CH₄ conversion and the constant diffusion model — the lowest. The Bosanquet approximation and unity Lewis number again have very close CH₄ conversion values with the lowest discrepancies around 1% for $\beta = 2$ and the highest 10.7% for

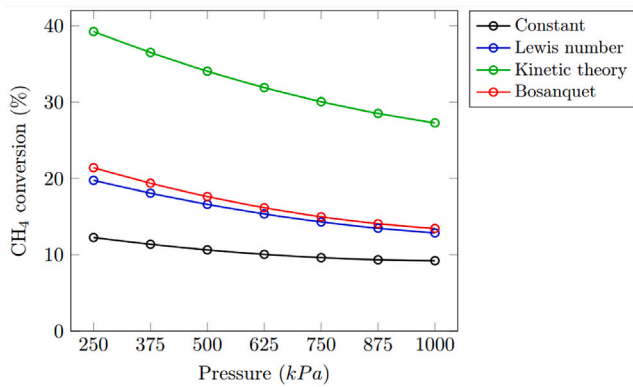


Fig. 9. Methane conversion dependence on operating pressure for $T_{wall} = 948$ K; residence time of 9000 kg s/kmol $_{CH_4}$; $H_2O:CH_4 = 2:1$ and particle diameter of 10 mm.

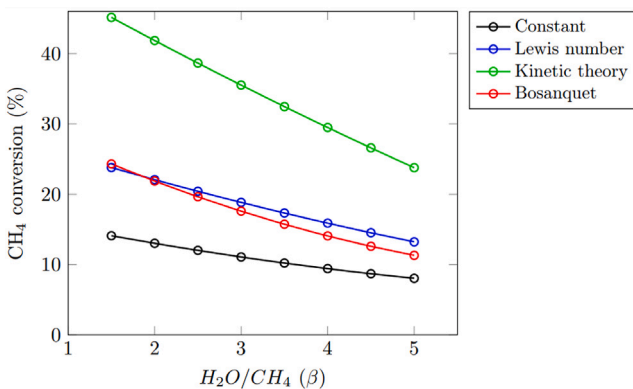


Fig. 10. Methane conversion dependence on H_2O/CH_4 ratio (β) for $T_{wall} = 948$ K; residence time of 9000 kg s/kmol $_{CH_4}$; operating pressure 120 kPa and particle diameter of 10 mm.

$\beta = 5$. One of the main differences between the two mechanisms is that the Bosanquet approximation varies from species concentrations, and the unity Lewis number model does not. However, the unity Lewis number model predicts CH_4 conversion similarly to the Bosanquet approximation in the $\beta \approx 2$ range, a widely adopted condition for industrial steam methane reforming.

4. Conclusion

This manuscript aimed to evaluate the influence of the diffusion mechanism on steam methane reforming process (SMR). Four diffusion mechanisms, namely, constant diffusion coefficient, Lewis number correlation, kinetic theory, and Bosanquet approximation, were compared via the CFD modeling. The SMR process were studied inside a reactor with near-isothermal conditions filled with $Ni-Al_2O_3$ catalyst particles. The numerical model was validated against experimental results. The validation revealed that the diffusion mechanisms had negligible effect on CH_4 conversion for small catalyst particles (up to 0.15 mm). Further, the cases with various mixture residence times, particle diameter, operating temperature, operating pressure and β were investigated.

The results showed that all the diffusion mechanism predict CH_4 conversion with similar dependence characters, but quantitatively, they differ significantly. For all cases considered, the use of the kinetic theory model and constant diffusion model resulted in CH_4 conversion on average by two times higher and 1.5 times lower than the Bosanquet approximation, respectively. The constant diffusion mechanism showed the lowest values and kinetic theory showed the highest values. In most cases, the Lewis number and Bosanquet approximation had very similar

predictions of CH_4 conversion. Most significant deviations (around 7%) between these two models were observed for temperatures above 1000K and $\beta > 3$. For other cases, both the Bosanquet approximation and Lewis number diffusion models yielded very similar CH_4 conversion. It should be noted that the Lewis number diffusion model requires less computational power and may be used in many cases. These results show the significance of the choice of the diffusion mechanism for the numerical simulation of steam methane reforming for real catalyst particles used for industrial application.

CRedit authorship contribution statement

Igor Karpilov: Formal analysis, Investigation, Methodology, Software, Validation, Writing – original draft. **Viacheslav Papkov:** Formal analysis, Investigation, Methodology, Software, Visualization. **Dmitry Pashchenko:** Conceptualization, Data curation, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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